 

# SMITH FIELD, AR, USA

WMO: **723443**

Lat: **36.191N** Lon: **94.491W** Elev: **1193** StdP: **14.07** Time Zone: **-6.00 (NAC)** Period: **94-19** WBAN: **53955**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **1** | **10.3** | **16.2** | **0.4** | **5.9** | **13.2** | **5.0** | **7.4** | **18.9** | **26.6** | **48.8** | **24.5** | **49.9** | **7.3** | **320** | **0.474** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **20.0** | **97.0** | **74.6** | **93.0** | **74.7** | **90.5** | **74.4** | **77.7** | **90.2** | **76.6** | **88.8** | **75.6** | **87.4** | **8.2** | **190** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **73.5** | **130.2** | **84.3** | **72.8** | **127.1** | **83.6** | **72.2** | **124.5** | **82.9** | **41.9** | **89.8** | **40.8** | **89.2** | **39.9** | **87.8** | **82.4** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **23.7** | **20.4** | **18.3** | DB | **4.3** | **99.6** | **5.3** | **4.9** | **0.5** | **103.1** | **-2.6** | **106.0** | **-5.6** | **108.8** | **-9.4** | **112.3** | (4) |
| (5) | | | | WB | **2.9** | **79.3** | **4.9** | **1.5** | **-0.6** | **80.4** | **-3.5** | **81.3** | **-6.2** | **82.2** | **-9.8** | **83.3** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | DBAvg | | **58.6** | **36.8** | **41.0** | **49.2** | **58.4** | **66.7** | **74.3** | **79.2** | **78.4** | **70.3** | **59.7** | **48.9** | **39.4** | (6) |
| (7) | | DBStd | | **17.09** | **11.29** | **11.31** | **10.60** | **8.85** | **7.21** | **5.35** | **4.92** | **5.28** | **7.19** | **8.70** | **9.48** | **10.19** | (7) |
| (8) | | HDD50 | | **1398** | **428** | **291** | **144** | **28** | **1** | **0** | **0** | **0** | **0** | **19** | **134** | **353** | (8) |
| (9) | | HDD65 | | **3869** | **873** | **673** | **497** | **229** | **70** | **5** | **0** | **1** | **32** | **208** | **488** | **793** | (9) |
| (10) | | CDD50 | | **4550** | **21** | **39** | **119** | **281** | **518** | **730** | **906** | **881** | **610** | **321** | **99** | **25** | (10) |
| (11) | | CDD65 | | **1542** | **0** | **1** | **6** | **32** | **122** | **285** | **441** | **417** | **191** | **44** | **3** | **0** | (11) |
| (12) | | CDH74 | | **14679** | **0** | **6** | **35** | **214** | **894** | **2584** | **4687** | **4248** | **1640** | **353** | **18** | **0** | (12) |
| (13) | | CDH80 | | **6165** | **0** | **1** | **2** | **31** | **209** | **954** | **2247** | **2049** | **599** | **72** | **1** | **0** | (13) |
| (14) | Wind | WSAvg | | **7.9** | **8.8** | **9.1** | **9.7** | **10.0** | **8.0** | **7.1** | **6.2** | **5.9** | **6.3** | **7.2** | **8.4** | **8.5** | (14) |
| (15) | Precipitation | PrecAvg | | **46.0** | **2.6** | **2.3** | **3.7** | **4.6** | **5.9** | **4.6** | **3.8** | **3.0** | **4.3** | **3.9** | **3.8** | **3.1** | (15) |
| (16) | | PrecMax | | **67.0** | **6.8** | **6.2** | **9.3** | **12.4** | **11.5** | **12.3** | **10.7** | **5.7** | **10.6** | **9.9** | **9.7** | **12.5** | (16) |
| (17) | | PrecMin | | **32.5** | **0.2** | **0.4** | **0.7** | **1.8** | **2.2** | **1.1** | **1.1** | **0.3** | **0.4** | **1.3** | **0.5** | **0.4** | (17) |
| (18) | | PrecStd | | **8.6** | **1.8** | **1.6** | **1.9** | **2.5** | **2.2** | **2.6** | **2.2** | **1.5** | **2.2** | **1.9** | **2.8** | **2.3** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **68.2** | **72.6** | **78.7** | **83.7** | **88.4** | **95.1** | **100.2** | **100.4** | **96.5** | **85.9** | **76.8** | **69.2** | (19) |
| (20) | | | MCWB | | **57.3** | **59.4** | **62.7** | **66.4** | **71.7** | **75.6** | **73.7** | **74.5** | **72.8** | **69.8** | **62.2** | **60.3** | (20) |
| (21) | | 2% | DB | | **63.6** | **66.5** | **73.1** | **79.4** | **84.3** | **90.8** | **96.9** | **97.2** | **90.2** | **81.7** | **71.9** | **64.1** | (21) |
| (22) | | | MCWB | | **54.8** | **56.4** | **59.9** | **64.0** | **70.9** | **74.8** | **75.1** | **74.2** | **72.0** | **66.3** | **60.0** | **56.6** | (22) |
| (23) | | 5% | DB | | **59.2** | **63.3** | **70.0** | **75.4** | **81.9** | **88.3** | **93.0** | **93.1** | **87.4** | **78.6** | **67.8** | **60.9** | (23) |
| (24) | | | MCWB | | **51.9** | **54.0** | **57.6** | **62.0** | **69.6** | **74.1** | **75.4** | **74.2** | **71.5** | **64.3** | **58.9** | **53.3** | (24) |
| (25) | | 10% | DB | | **54.6** | **59.3** | **65.8** | **72.7** | **79.3** | **85.6** | **90.5** | **90.4** | **82.5** | **73.4** | **64.1** | **55.2** | (25) |
| (26) | | | MCWB | | **47.3** | **51.0** | **56.0** | **60.6** | **67.7** | **73.1** | **75.4** | **74.0** | **69.3** | **62.7** | **55.8** | **48.4** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **60.2** | **62.5** | **65.1** | **69.0** | **74.9** | **78.2** | **79.3** | **79.2** | **75.9** | **71.6** | **65.9** | **62.2** | (27) |
| (28) | | | MCDB | | **66.1** | **69.6** | **75.4** | **79.4** | **84.0** | **89.9** | **89.9** | **91.5** | **87.7** | **81.6** | **72.6** | **67.4** | (28) |
| (29) | | 2% | WB | | **56.7** | **59.0** | **62.1** | **66.6** | **73.0** | **76.6** | **77.8** | **77.5** | **74.6** | **69.0** | **62.6** | **58.3** | (29) |
| (30) | | | MCDB | | **61.6** | **64.8** | **71.1** | **76.3** | **82.2** | **87.8** | **90.2** | **90.4** | **85.7** | **78.4** | **68.6** | **63.5** | (30) |
| (31) | | 5% | WB | | **52.6** | **55.7** | **59.6** | **64.5** | **71.1** | **75.2** | **76.8** | **76.2** | **73.0** | **66.8** | **59.9** | **54.3** | (31) |
| (32) | | | MCDB | | **57.9** | **61.8** | **67.0** | **73.0** | **79.7** | **85.3** | **89.3** | **88.9** | **83.6** | **74.8** | **66.0** | **59.2** | (32) |
| (33) | | 10% | WB | | **47.5** | **51.5** | **56.9** | **62.8** | **69.1** | **73.9** | **75.8** | **75.1** | **71.2** | **64.4** | **57.1** | **50.2** | (33) |
| (34) | | | MCDB | | **54.2** | **58.4** | **64.8** | **71.1** | **77.8** | **83.8** | **88.1** | **87.4** | **80.4** | **72.3** | **63.0** | **55.0** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **18.8** | **19.4** | **20.0** | **20.8** | **19.1** | **18.9** | **20.0** | **20.5** | **20.2** | **21.8** | **19.7** | **17.8** | (35) |
| (36) | | 5% DB | MCDBR | | **23.9** | **24.0** | **23.7** | **22.8** | **20.8** | **21.6** | **23.3** | **24.9** | **23.5** | **24.4** | **23.1** | **23.6** | (36) |
| (37) | | | MCWBR | | **17.0** | **17.1** | **15.0** | **13.7** | **11.4** | **10.1** | **8.8** | **9.1** | **10.5** | **13.2** | **15.5** | **17.2** | (37) |
| (38) | | 5% WB | MCDBR | | **20.2** | **20.9** | **20.0** | **19.3** | **18.4** | **19.0** | **20.5** | **20.6** | **19.2** | **20.0** | **20.4** | **19.9** | (38) |
| (39) | | | MCWBR | | **16.3** | **16.7** | **14.5** | **13.4** | **10.9** | **10.2** | **9.2** | **8.9** | **9.8** | **12.3** | **16.0** | **16.4** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.303** | **0.306** | **0.342** | **0.379** | **0.422** | **0.424** | **0.443** | **0.444** | **0.388** | **0.345** | **0.313** | **0.292** | (40) |
| (41) | | taud | | | **2.486** | **2.488** | **2.418** | **2.323** | **2.256** | **2.277** | **2.274** | **2.247** | **2.372** | **2.479** | **2.520** | **2.533** | (41) |
| (42) | | Ebn at Noon | | | **285** | **298** | **294** | **286** | **274** | **273** | **266** | **264** | **274** | **279** | **279** | **282** | (42) |
| (43) | | Edh at Noon | | | **26** | **29** | **35** | **40** | **43** | **43** | **43** | **43** | **36** | **29** | **25** | **23** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **781** | **948** | **1277** | **1607** | **1784** | **2039** | **2032** | **1842** | **1570** | **1166** | **842** | **666** | (44) |
| (45) | | RadStd | | | **71** | **135** | **76** | **93** | **154** | **187** | **124** | **123** | **116** | **152** | **94** | **85** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD50 | HDD65 | CDD50 | CDD65 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (29 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

[Nomenclature: See separate page]